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Shift system for a vehicle

Abstract

A shift system includes a shifter including a first switch, a second switch, and an interlock switch and being operable between a plurality of gear states. A controller is communicatively coupled with the shifter to receive signals that correspond to each of the first switch, the second switch, and the interlock switch. The controller includes a diagnostic counter that includes a passing criteria and a predetermined diagnostic threshold and is configured to adjust the diagnostic counter based on at least one of the signals received from the shifter being inconsistent with the others of the signals received.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2861464	12/1957	Jensen	74/335	G05D 3/127
4611286	12/1985	Nishimura	902/22	G07D 11/00
11892076	12/2023	Kim	N/A	F16H 61/18
2017/0248227	12/2016	Zhao	N/A	F16H 61/36

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Background/Summary

INTRODUCTION

- (1) The information provided in this section is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against present disclosure.
- (2) The present disclosure relates generally to a shift system for a vehicle.
- (3) A vehicle lever shifter is used in conventional vehicles to transition the vehicle between different operative states. For example, a driver may select one of the levers to put the vehicle in drive. While the vehicle operative states corresponding to movement are operable via the lever, conventional shift systems have a separate button or switch for a parking function. To communicate the selection of an operative state, the lever shift uses two channels to communicate a position of one of the levers or park button of a vehicle to a vehicle module. The vehicle module may function to execute the lever selection through various methods.
- (4) For functional reasons, three communication channels would provide increased position communication between the lever shifter and the vehicle module. However, vehicle modules typically only have two available channels for lever shifter communication. The communication channels may be configured using a serial communication bus between the vehicle lever shifter and the vehicle module. The signals sent back and forth between the lever shifter and the vehicle module communicate the selected operative state of the vehicle with each selection using multiple signals to confirm the selection.

SUMMARY

- (5) In one configuration, a vehicle shift system includes a first gear switch operable between a plurality of gear states, a second gear switch operable between the plurality of gear states, and an interlock switch operable between an open state and a closed state. The closed state corresponds to one of the plurality of gear states. A controller is communicatively coupled to each of the first gear switch, the second gear switch, and the interlock switch. The controller includes a diagnostic counter and is configured to register a diagnostic on the diagnostic counter when a signal from at least one of the first gear switch, the second gear switch, and the interlock switch is different from

the other of the first gear switch, the second gear switch, and the interlock switch.

(6) The system may include one or more of the following optional features. For example, the plurality of gear states may include a reverse state, a neutral state, and a drive state. The closed state of the interlock switch may correspond to activation of one of the reverse state, the neutral state, and the drive state. The controller may include a memory storing a predetermined diagnostic threshold and may be configured to compare a signal from the interlock switch with a signal from each of the first gear switch and the second gear switch, respectively. The controller may be configured to detect one or more of the first gear switch, the second gear switch, and the interlock switch being activated via the respective signals. The diagnostic counter of the controller may be configured with a passing criteria and a non-passing criteria and may be configured to reduce a diagnostic log based on a predetermined number of pass accruals of the passing criteria. The pass accruals may include the controller receiving a signal from each of the first gear switch, the second gear switch, and the interlock switch.

(7) In another configuration, a shift system includes a shifter including a first switch, a second switch, and an interlock switch and being operable between a plurality of gear states. A controller is communicatively coupled with the shifter to receive signals that correspond to each of the first switch, the second switch, and the interlock switch. The controller includes a diagnostic counter that includes a passing criteria and a predetermined diagnostic threshold and is configured to adjust the diagnostic counter based on at least one of the signals received from the shifter being inconsistent with the others of the signals received.

(8) In some examples, the interlock switch may be a resistor ladder circuit operable between an open state and a closed state. The controller may be configured to add a diagnostic to the diagnostic counter when a signal of the interlock switch is different than a signal of at least one of the first switch and the second switch. The first switch and the second switch may each include a plurality of gear states that include a reverse state, a neutral state, and a drive state. The closed state of the interlock switch may include one of the plurality of gear states. Optionally, the controller may be configured to add a diagnostic to the diagnostic counter when the interlock switch is in the open state and the first switch and the second switch are in a same one of the plurality of gears. In some aspects, the diagnostic counter may include a standard diagnostic log and an interlock diagnostic log. The standard diagnostic log may correspond to signals received from the first switch and the second switch, and the interlock diagnostic log may correspond to a signal received from the interlock switch.

(9) In yet another configuration, a shift system for a vehicle includes a gear switch including a first switch and a second switch each operable between a reverse state, a neutral state, and a drive state. An interlock switch is communicatively coupled to the gear switch and is operable between an open state and a closed state selectively corresponding to one of the reverse state, the neutral state, and the drive state. A controller is communicatively coupled to the gear switch and the interlock switch and is configured to receive signals from the interlock switch and the gear switch. The controller stores a diagnostic counter that is configured to track a difference between the signal received from the interlock switch and the signal received from the gear switch. The diagnostic counter includes a predetermined diagnostic threshold for each of the gear switch and the interlock switch, respectively.

(10) In some examples, the signal of the interlock switch may include one of the open state and the closed state of the interlock switch and the signals of the gear switch may include one of the reverse state, the neutral state, and the drive state of each of the first switch and the second switch. The controller may be configured with a passing criteria and may be configured to reduce a diagnostic count of the diagnostic counter based on the passing criteria. The passing criteria may be defined by the interlock switch being in the closed state and both of the first switch and the second switch being in a same state of any one of the reverse state, the neutral state, and the drive state. In some aspects, the controller may be configured to determine an activation time between the signals

received from the gear switch and the interlock switch. The controller may be configured to log a pass accrual based on the activation time.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The drawings described herein are for illustrative purposes only of selected configurations and are not intended to limit the scope of the present disclosure.
- (2) FIG. 1 is a perspective view of a vehicle according to the present disclosure;
- (3) FIG. 2 is an enlarged partial perspective view of a shifter driver interface according to the present disclosure;
- (4) FIG. 3 is a functional block diagram of an example of a shift system providing communication between a shifter driver interface, a gear switch, an interlock switch, and a controller according to the present disclosure;
- (5) FIG. 4 is a functional block diagram of an example resistor ladder for an interlock switch according to the present disclosure;
- (6) FIG. 5 is a functional block diagram of an example diagnostic counter according to the present disclosure; and
- (7) FIG. 6 is an example method of determining a diagnostic for a shifter driver interface according to the present disclosure.
- (8) Corresponding reference numerals indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION

- (9) Example configurations will now be described more fully with reference to the accompanying drawings. Example configurations are provided so that this disclosure will be thorough, and will fully convey the scope of the disclosure to those of ordinary skill in the art. Specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of configurations of the present disclosure. It will be apparent to those of ordinary skill in the art that specific details need not be employed, that example configurations may be embodied in many different forms, and that the specific details and the example configurations should not be construed to limit the scope of the disclosure.
- (10) The terminology used herein is for the purpose of describing particular exemplary configurations only and is not intended to be limiting. As used herein, the singular articles “a,” “an,” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “comprising,” “including,” and “having,” are inclusive and therefore specify the presence of features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. Additional or alternative steps may be employed.
- (11) When an element or layer is referred to as being “on,” “engaged to,” “connected to,” “attached to,” or “coupled to” another element or layer, it may be directly on, engaged, connected, attached, or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to,” “directly connected to,” “directly attached to,” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(12) The terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections. These elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as “first,” “second,” and other numerical terms do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example configurations.

(13) In this application, including the definitions below; the term module may be replaced with the term circuit. The term module may refer to, be part of, or include an Application Specific Integrated Circuit (ASIC); a digital, analog, or mixed analog/digital discrete circuit; a digital, analog, or mixed analog/digital integrated circuit; a combinational logic circuit; a field programmable gate array (FPGA); a processor (shared, dedicated, or group) that executes code; memory (shared, dedicated, or group) that stores code executed by a processor; other suitable hardware components that provide the described functionality; or a combination of some or all of the above, such as in a system-on-chip.

(14) The term code, as used above, may include software, firmware, and/or microcode, and may refer to programs, routines, functions, classes, and/or objects. The term shared processor encompasses a single processor that executes some or all code from multiple modules. The term group processor encompasses a processor that, in combination with additional processors, executes some or all code from one or more modules. The term shared memory encompasses a single memory that stores some or all code from multiple modules. The term group memory encompasses a memory that, in combination with additional memories, stores some or all code from one or more modules. The term memory may be a subset of the term computer-readable medium. The term computer-readable medium does not encompass transitory electrical and electromagnetic signals propagating through a medium, and may therefore be considered tangible and non-transitory memory. Non-limiting examples of a non-transitory memory include a tangible computer readable medium including a nonvolatile memory, magnetic storage, and optical storage.

(15) The apparatuses and methods described in this application may be partially or fully implemented by one or more computer programs executed by one or more processors. The computer programs include processor-executable instructions that are stored on at least one non-transitory tangible computer readable medium. The computer programs may also include and/or rely on stored data.

(16) A software application (i.e., a software resource) may refer to computer software that causes a computing device to perform a task. In some examples, a software application may be referred to as an “application,” an “app,” or a “program.” Example applications include, but are not limited to, system diagnostic applications, system management applications, system maintenance applications, word processing applications, spreadsheet applications, messaging applications, media streaming applications, social networking applications, and gaming applications.

(17) The non-transitory memory may be physical devices used to store programs (e.g., sequences of instructions) or data (e.g., program state information) on a temporary or permanent basis for use by a computing device. The non-transitory memory may be volatile and/or non-volatile addressable semiconductor memory. Examples of non-volatile memory include, but are not limited to, flash memory and read-only memory (ROM)/programmable read-only memory (PROM)/erasable programmable read-only memory (EPROM)/electronically erasable programmable read-only memory (EEPROM) (e.g., typically used for firmware, such as boot programs). Examples of volatile memory include, but are not limited to, random access memory (RAM), dynamic random access memory (DRAM), static random access memory (SRAM), phase change memory (PCM) as well as disks or tapes.

(18) These computer programs (also known as programs, software, software applications or code)

include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms “machine-readable medium” and “computer-readable medium” refer to any computer program product, non-transitory computer readable medium, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor.

(19) Various implementations of the systems and techniques described herein can be realized in digital electronic and/or optical circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof. These various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device.

(20) The processes and logic flows described in this specification can be performed by one or more programmable processors, also referred to as data processing hardware, executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by special purpose logic circuitry. e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit). Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read only memory or a random access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data. e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices. e.g., EPROM, EEPROM, and flash memory devices; magnetic disks. e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

(21) To provide for interaction with a user, one or more aspects of the disclosure can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube), LCD (liquid crystal display) monitor, or touch screen for displaying information to the user and optionally a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input. In addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; for example, by sending web pages to a web browser on a user's client device in response to requests received from the web browser.

(22) Referring to FIGS. 1-4, a vehicle **10** includes a vehicle body **12**, which defines an interior cabin **14** of the vehicle **10** with one or more seating rows **16** including a driver seat **16a** and a passenger seat **16b**. A display console **18** may be positioned between the driver seat **16a** and the passenger seat **16b** to display information to a driver or occupant of the vehicle **10** including, but

not limited to, navigation, entertainment, and vehicle service information. The vehicle **10** includes a shifter driver interface **20** communicatively coupled with a vehicle propulsion or shift system **100** of the vehicle **10**. The shifter driver interface **20** is operable to adjust gears of the vehicle **10** to translate between various operative states, as described below.

(23) A controller **200** of the vehicle **10** is communicatively coupled to the shift system **100** and the display console **18** to display information to the driver from the controller **200**. For example, the display console **18** may display notifications from the controller **200** pertaining to the shifter driver interface **20**, as described below, notifying the driver of potential service recommendations for the vehicle **10**. In some configurations, the controller **200** is configured as part of the shift system **100** and is communicatively coupled to individual elements of the shift system **100**.

(24) As illustrated in FIG. 2, the shifter driver interface **20** includes a plurality of actuators **22** electrically coupled to the shift system **100** and configured to select one of a plurality of gear states **24** of the vehicle **10**. The plurality of actuators **22** includes, but is not limited to, a drive actuator **22a**, a reverse actuator **22b**, and a neutral actuator **22c** each respectively corresponding to one of the plurality of gear states **24**. For example, the gear states **24** include a drive state **24a**, a reverse state **24b**, and a neutral state **24c** of the vehicle **10**. The gear states **24** of the vehicle **10** are defined by a propulsion activity (e.g., drive, reverse, neutral) of the vehicle **10**. The shifter driver interface **20** activates the propulsion activity in response to an input to one of the actuators **22a-22c**.

(25) With further reference to FIGS. 1-4, the shift system **100** of the shifter driver interface **20** includes a gear shifter or switch **102** and an interlock switch **104**, each in communication with the plurality of actuators **22**. The gear switch **102** is communicatively coupled to the controller **200** and includes a first switch **102a** and a second switch **102b**. As used herein, the term gear switch **102** may collectively refer to the first switch **102a** and the second switch **102b**. The gear switch **102** transmits gear signals **106a**, **106b** to the controller **200** corresponding to one of the gear states **24** via both of the first switch **102a** and the second switch **102b**, respectively. For example, the first switch **102a** transmits a first gear signal **106a** to the controller **200**, and the second switch **102b** transmits a second gear signal **106b** to the controller **200**.

(26) Both the first gear signal **106a** and the second gear signal **106b** correspond to one or more of the gear states **24**, mentioned above, to indicate to the controller **200** the selection of one or more of the actuators **22a-22c** by the driver. For example, the driver may actuate the drive actuator **22a**, and both the first switch **102a** and the second switch **102b** are configured to send respective first and second gear signals **106a**, **106b** to the controller **200** indicating the selection of the drive state **24a**. Stated differently, the gear signals **106a**, **106b** transmit the selection of one of the drive state **24a**, the reverse state **24b**, and the neutral state **24c** to the controller **200** via the corresponding actuator **22a-22c** by the driver. The gear signals **106a**, **106b** are specific to the gear state **24a-24c**, such that the controller **200** identifies the selected gear state **24a-24c** upon receipt of the gear signals **106a**, **106b**. In some examples, each of the gear signals **106a**, **106b** may output a voltage value received by the controller **200**. For example, one or more of the gear signals **106a**, **106b** may include a voltage that is perceived by the controller **200** to indicate the pressing of the respective actuator **24a-24c**. Thus, the controller **200** may identify whether multiple actuators **24a-24c** have been pressed or otherwise actuated simultaneously. Namely, the shift system **100** controls or otherwise executes shifting of the shifter driver interface **20** between various transmission states corresponding to vehicle propulsion and communicates the shifting election with the controller **200**.

(27) Referring still to FIGS. 1-4, the first switch **102a** is communicatively coupled to the controller **200** to transmit the first gear signal **106a** corresponding to one or more of the drive state **24a**, the reverse state **24b**, and the neutral state **24c**. The first gear signal **106a** is received by the controller **200** with one or more respective states **24a-24c** and is registered by the controller **200** as a press function. The second switch **102b** is communicatively coupled to the controller **200** to transmit the second gear signal **106b** corresponding to the same one or more of the drive state **24a**, the reverse

state **24b**, and the neutral state **24c** as the first gear signal **106a**. If the second gear signal **106b** is different than the first gear signal **106a**, then the controller **200** will log a diagnostic **202**, described in more detail below. Each of the first gear signal **106a** and the second gear signal **106b** is compared with an interlock signal **110** received by the controller **200** from the interlock switch **104**. (28) The interlock switch **104** is operable between an open state **110a** and a closed state **110b**, collectively communicated to the controller **200** as the interlock signal **110**. Similar to the gear switch **102**, the interlock switch **104** is configured to detect actuation of one of the actuators **22a-22c** by the driver. The interlock switch **104**, like the gear switch **102**, detects each of the drive state **24a**, the reverse state **24b**, and the neutral state **24c**. Unlike the gear switch **102**, the interlock switch **104** transmits a binary, interlock signal **110** to the controller **200**. Stated differently, the interlock switch **104** communicates either the open state **110a** or the closed state **110b** with the controller **200** irrespective of the specific gear state **24** selected. For example, the interlock signal **110** does not transmit any information or identifying data as to which of the gear states **24** was selected.

(29) While the interlock switch **104** detects the selected gear state **24**, the interlock switch **104** transmits the binary interlock signal **110** to the controller **200**. The interlock signal **110**, as described in more detail below; is used as a comparison check by the controller **200** with the gear signals **106a**, **106b**. The binary configuration of the interlock signal **110** advantageously reduces manufacturing costs associated with the shift system **100** by minimizing the number of communication buses used in the shift system **100**. For example, the gear switch **102** utilizes multiple communication pathways and buses in association with the gear states **24** to translate the detected state **24a-24c** to the controller **200**. Comparatively, the interlock switch **104** is configured on a single resistor ladder circuit **112** that detects the selection of one of the gear states **24** as a binary open state **110a** or closed state **110b** of the interlock switch **104**. It is advantageous for the shift system **100** to include both the gear switch **102** that identifies the specific gear state **24** selected by the driver and the interlock switch **104** as backup confirmation that one of the gear states **24** was selected, irrespective of which one.

(30) With reference now to FIGS. 3-5, the controller **200** is configured to receive the gear signals **106a**, **106b** identifying one of the gear states **24** from each of the first switch **102a** and the second switch **102b**. The controller **200** also receives the interlock signal **110** from the interlock switch **104** indicating the state **110a**, **110b** of the interlock switch **104**. As mentioned above, the state **110a**, **110b** of the interlock switch **104** corresponds to one of the open state **110a** and the closed state **110b**. The closed state **110b** corresponds to a pressed condition of one or more of the actuators **22a-22c**, and the open state **110a** corresponds to an unpressed condition of all of the actuators **22a-22c**, described in more detail below.

(31) With specific reference to FIG. 4, the interlock switch **104** serves as a confirmation switch to validate the information received from the gear switch **102** to the controller **200**. In some aspects, the interlock switch **104** may confirm a failure of one of the gear switches **102**. The interlock switch **104** is mapped onto a single resistor ladder circuit **112** spanning the interlock switch **104** and the controller **200**. As illustrated, the interlock switch **104** includes three input switches **124a**, **124b**, **124c** corresponding to the respective gear states **24** and operable between the open state **110a** and the closed state **110b**. As described further below, one of the input switches **124a-124c** moves from the open state **110a** to the closed state **110b** in response to an input on a corresponding actuator **22a-22c**. In response, the interlock switch **104** communicates the interlock signal **110** to the controller **200** of the closed state **110b**. The interlock signal **110** is free from identifying information associated with the respective input switch **124a-124c**, such that the controller **200** identifies the state of the interlock switch **104** and not the gear state **24** from the interlock signal **110**. The controller **200** may then use the state **110a**, **110b** of the interlock switch **104** to validate the gear signals **106a**, **106b** received from the gear switch **102**.

(32) The state **110a**, **110b** of the interlock switch **104** may be determined by the controller **200**

based on a voltage reference **114** of the interlock signal **110**. For example, the resistor ladder circuit **112** may utilize an analog-to-digital (A/D) converter **116** that receives the interlock signal **110** to convert the interlock signal **110** to the binary signal read by the controller **200**. The voltage reference **114** is presented to the A/D converter **116** based on the state **110a**, **110b** of the interlock signal **110** and may be used in converting the binary signal **110** for the controller **200**. Using the analog voltage reference **114** provides cost saving advantages as compared to the incorporation of a third communication pathway with a similar construction as the communication pathways used for the gear switches **102**.

(33) Referring to FIGS. 3-6, the controller **200** includes a memory **204** that stores a predetermined diagnostic threshold **206**. The predetermined diagnostic threshold **206** is a threshold value of the diagnostics **202** accrued during operation of the vehicle **10**. The controller **200** is configured to prevent the shifter driver interface **20** from executing the selected gear state **24** at the predetermined diagnostic threshold **206** and may issue a notification indicating recommended service to the shifter driver interface **20**. Stated differently, the predetermined diagnostic threshold **206** corresponds to an accrual value at which the controller **200** is configured to prevent gear shifting of the shifter driver interface **20**. The controller **200** may, in some aspects, send a notification to the driver that service or maintenance is recommended on the vehicle **10** to address the diagnostics **202**.

(34) The predetermined diagnostic threshold **206** includes a standard threshold **208** that sets the threshold value of diagnostics **202** for the gear switch **102**, with each of the first switch **102a** and the second switch **102b** having the same standard threshold **208**. Additionally or alternatively, the standard threshold **208** may be further segmented, such that the first switch **102a** and the second switch **102b** may have different thresholds. The diagnostic threshold **206** also includes an interlock threshold **210** corresponding to the interlock switch **104**. The interlock threshold **210** sets the threshold number of diagnostics **202** specifically related to the interlock signal **110**. For example, the controller **200** may similarly prevent actuation of the gear state when the interlock switch **104** meets or exceeds the interlock threshold **210**. As noted above, the controller **200** may also issue a notification to the driver of recommended service to the shifter driver interface **20** based on either or both of the standard threshold **208** and the interlock diagnostic log **210** being met.

(35) The controller **200** includes a diagnostic counter **212** configured with the diagnostics **202** to track the signals **106a**, **106b**, **110** received from the gear switch **102** and the interlock switch **104**. The diagnostic counter **212** may, in combination with functions of the controller **200**, record a discrepancy whenever there is a discrepancy between any of the signals **106a**, **106b**, **110**. Where a discrepancy occurs, the diagnostic counter **212** will log a diagnostic **202** associated with the inconsistent switch **102**, **104**. The diagnostic **202** includes a standard diagnostic log **214** and an interlock diagnostic log **216**. The diagnostics **202** may be categorized with respect to the standard diagnostic log **214** and the interlock diagnostic log **216** based on the signal received by the controller **200**.

(36) The standard diagnostic log **214** tracks the number of diagnostics **202** that accrue over a lifetime of the vehicle **10**. Similarly, the interlock diagnostic log **216** tracks the number of diagnostics **202** that accrue for the interlock switch **104** over the lifetime of the vehicle **10**. In some examples, the diagnostic counter **212** may be manually cleared using a servicing tool to reset the diagnostic counter **212**. In addition to logging diagnostics **202**, the diagnostic counter **212** may also log instances where the signals **106a**, **106b**, **110** are all validated, as described below with respect to FIG. 5.

(37) With continued reference to FIGS. 3-6, the controller **200** may receive the interlock signal **110** and determine an inconsistency as compared with the gear signals **106a**, **106b**. Thus, the diagnostic counter **212** may accrue a diagnostic **202** on the interlock diagnostic log **216**. Examples of the process of accruing a diagnostic **202** are described in more detail below. The accruals associated with diagnostics **202** may be referred to as diagnostic accruals **220**. It is also contemplated that the

diagnostic counter **212** may also record pass accruals **222** for each of the gear switch **102** and the interlock switch **104**. The pass accruals **222** mark the number of times all three of the switches **102a**, **102b**, **104** were registered by the controller **200**. Stated differently, the pass accruals **222** may tabulate each time the gear signals **106a**, **106b** are confirmed by the interlock signal **110**.

(38) The diagnostic accruals **220** and pass accruals **222** may be incorporated as passing criteria and non-passing criteria stored in the memory **204**. The controller **200** may utilize the pass accruals **222** in determining whether to activate the shifter driver interface **20** or prompt the driver with a notification for recommended service to the shifter driver interface **20**. The number of pass accruals **222** may nullify or incrementally reduce the diagnostic accruals **220**. Similarly, the controller **200** may reduce the diagnostic logs **214**, **216** based on a predetermined number of pass accruals **222** that correspond to the passing criteria. Each of the pass accruals **222** and the diagnostic accruals **220** are tabulated for each signaled state **24a-24c** from the first switch **102a** and the second switch **102b**. For example, each of the gear states **24** are represented for each of the pass accruals **222** and the diagnostic accruals **220**, such that each gear state **24** may have a different value with respect to each of the pass accruals **222** and the diagnostic accruals **220**. In some examples, the drive state **24a** may have more pass accruals on the standard diagnostic log **214** as compared to the neutral state **24c**, which results in more diagnostics **202** being logged for the drive state **24a**.

(39) As illustrated in FIG. 5, the standard diagnostic log **214** is depicted with a first switch profile **224a** and a second switch profile **224b**, each with a respective counter for the detected gear states **24**. The controller **200** is configured to receive the signals **106a**, **106b** from each of the first switch **102a** and the second switch **102b** corresponding to one of the three gear states **24**. A potential diagnostic **202** may be associated with the selected gear state **24** depending on what is received by the controller **200**. For example, the controller **200** may receive the first signal **106a** corresponding to the drive state **24a** but does not receive the second signal **106b** from the second switch **102b**. In this example, the diagnostic counter **212** would record a diagnostic **202** with the second switch profile **224b** of the standard diagnostic log **214**, which adds to the diagnostic accrual **220** of the drive state **24a** associated with the second switch profile **224b**. The interlock diagnostic log **216**, as mentioned above, is based upon the state **110a**, **110b** of the interlock switch **104** when compared to the gear switch **102**. Thus, the controller **200** may identify both the switches **102**, **104** having repeated diagnostics **202** logged, but may also identify the selected state resulting in the diagnostic **202**.

(40) The diagnostic accrual **220** is added incrementally based on the respective signals **110**, **106a**, **106b** received from the interlock switch **104** and the gear switch **102** and is evaluated by the controller **200**. In some examples, the gear signals **106a**, **106b** from the gear switch **102** may both identify the reverse state **24b** and the interlock signal **110** from the interlock switch **104** identifies the open state **110a**. Thus, a diagnostic **202** would accrue to the interlock diagnostic log **216** as the interlock signal **110** was determined to be inconsistent with both of the signals **106a**, **106b** from the gear switch **102**. Stated differently, the pass accrual **222** for the interlock switch **104** occurs when all of the gear signals **106a**, **106b** correspond to the interlock signal **110** in that both of the signals **106a**, **106b** indicates selection of one of the gear states **24** and corresponds with the closed state **110b** of the interlock switch **104**. In a passing criteria, the user may select one of the gear states **24**, which is communicated to the controller **200** via the gear signals **106a**, **106b** from the first switch **102a** and the second switch **102b**, respectively, and is also communicated to the controller **200** as the interlock signal **110** in the closed state **110b** of the interlock signal **110** from the interlock switch **104**.

(41) As noted above, the interlock switch **104** is a binary switch and, thus, sends a communication corresponding to either the open state **110a** or the closed state **110b** without identifying the selected gear state **24**. The open state **110a** and the closed state **110b** both relate to each one of the gear states **24** while being free from identification of a specific one of the gear states **24**. Comparatively, the gear signals **106a**, **106b** sent from the gear switch **102** are specific to and identify the gear state

24 selected, such that the controller **200** receives at least one of three signal options from the gear switch **102**. The gear signals **106a**, **106b** from the gear switch **102** therefore include at least one of the drive state **24a**, the reverse state **24b**, and the neutral state **24c**.

(42) Referring still to FIGS. 3-6, the controller **200** compares the gear signals **106a**, **106b** from the gear switch **102** with the interlock signal **110** to determine whether the gear switch **102** and the interlock switch **104** are cooperatively activated. Upon comparison of the gear signals **106a**, **106b** with the interlock signal **110**, the controller **200** determines whether to assign a diagnostic **202** to one or both of the interlock diagnostic log **216** and the standard diagnostic log **214**. Each diagnostic **202** assignment accrues on the respective diagnostic log **214**, **216** until the predetermined diagnostic threshold **206** is met.

(43) As mentioned above, the diagnostic counter **212** is configured with the interlock diagnostic log **216** and the standard diagnostic log **214** to segregate the respective diagnostics **202** and more readily identify the source of the diagnostic **202**. Given that the interlock switch **104** is a binary switch as compared to the specified switches **102a**, **102b** of the gear switch **102**, the interlock switch **104** may have a different diagnostic count threshold as compared to the gear switch **102**. The standard diagnostic log **214**, for the first switch **102a** and the second switch **102b**, tracks the respective gear state **24** of each of the first signal **106a** and the second signal **106b**. As described above, the first signal **106a** and the second signal **106b** translate the selected gear state **24** and, thus, the controller **200** is able to identify which of the three gear states **24** may have a distinct issue as compared with the other gear states **24**. Comparatively, the interlock switch **104** is configured to give a shorthand identification of the state of the interlock switch **104**. Based on the binary configuration of the interlock signal **110**, an interlock diagnostic threshold **206** may be different than the standard diagnostic threshold **206**.

(44) With further reference to FIGS. 3-6, the shift system **100**, in operation, is configured to determine an inconsistency between each of the first switch **102a**, the second switch **102b**, and the interlock switch **104** during the selection of one of the drive state **24a**, the reverse state **24b**, and the neutral state **24c**. The inconsistency is logged in the diagnostic counter **212** of the controller **200** as a diagnostic **202**. The diagnostic **202** may be more specifically attributed to one of the standard diagnostic log **214** and the interlock diagnostic log **216** depending on which of the first switch **102a**, the second switch **102b**, and the interlock switch **104** is attributed to the diagnostic **202**. In some instances, more than one diagnostic **202** may be applied, such that both the standard diagnostic log **214** and the interlock diagnostic log **216** may accrue a diagnostic **202**. It is also contemplated that the controller **200** may assess an activation time **230** between receiving each of the signals **106a**, **106b**, **110** to identify a potential delay in activation between the gear switch **102** and the interlock switch **104**. If the activation time **230** is satisfied, then the controller **200** may log a pass accrual **222** and activate the selected gear state **24**. Comparatively, if the activation time **230** exceeds a threshold time, then the controller **200** can log a diagnostic **202** in the diagnostic log **214**, **216** for the respective delayed switch **102**, **104**.

(45) The diagnostics **202** are accrued on the diagnostic counter **212** until the predetermined diagnostic threshold **206** stored on the memory **204** is met. Once the predetermined diagnostic threshold **206** is met, the controller **200** may prevent activation of the shifter driver interface **20** via the respective switch **102**, **104** and may issue a notification to be displayed on the display **18** (FIG. 2) notifying the driver of recommended service to the shifter driver interface **20**. While the diagnostic counter **212** accrues diagnostics **202** over time, the diagnostic counter **212** may be manually reset to clear the diagnostic counter of accrued diagnostics **202**. Further, the controller **200** may activate the selected gear state **24** even if a diagnostic **202** is logged in the diagnostic counter **212**, so long as the diagnostic logs **214**, **216** are below the predetermined diagnostic threshold **206**.

(46) The controller **200** may activate the selected gear state **24** despite a diagnostic **202** being logged when at least two of the three switches **102a**, **102b**, **104** are registered by the controller **200**.

For example, the driver may select the neutral state **24c** via the neutral actuator **22c** (FIG. 2). In this example, the controller **200** may receive the first gear signal **106a** indicating the neutral state **24c** selection and the interlock signal **110** indicating the closed state **110b**, but does not receive a signal from the second switch **102b**. Additionally or alternatively, the second gear signal **106b** may be received by the controller **200** in a delayed manner outside of the activation time **230**, which may result in a diagnostic **202**. The controller **200** may then log a diagnostic **202** on the second switch profile **224b** corresponding to the neutral state **24c** and determine whether that diagnostic **202** exceeds the predetermined diagnostic threshold **206** for the neutral state of the second switch profile **224b**. If the diagnostic **202** does not exceed the predetermined diagnostic threshold **206**, then the controller **200** may activate the neutral state **24c** of the shifter driver interface **20**. A similar process may occur with any one of the gear switches **102** and interlock switch **104** with respective to any of the gear states **24**.

(47) With reference again to FIGS. 1-6, a method **300** of operation of the shift system **100** is provided. The method **300** of FIG. 6 is provided as an example, other similar methods may be implemented. At **302**, the controller **200** detects an input signal **106a**, **106b**, **110** from at least one of the gear switch **102** and the interlock switch **104** and, at **304**, the controller **200** determines whether three input signals **106a**, **106b**, **110** are detected. If the signals **106a**, **106b**, **110** are detected, then at **306**, the diagnostic counter **212** logs a pass accrual **222** and activates the switches **102**, **104**. If one or more of the signals **106a**, **106b**, **110** are not detected, then at **308**, the controller **200** identifies which of the signals **106a**, **106b**, **110** is not detected and logs a diagnostic **202** with the diagnostic counter **212**. The controller **200** may also identify which of the gear states **24** was associated with the diagnostic **202**. At **310**, the controller **200** determines if the diagnostic counter **212** is above the predetermined diagnostic threshold **206**. If the diagnostic counter **212** is not above the predetermined diagnostic threshold **206**, then at **312**, the controller **200** commands the vehicle **10** to activate the switches **102**, **104** corresponding to the gear request. Comparatively, if the diagnostic counter **212** exceeds the predetermined diagnostic threshold **206**, then the diagnostic **202** is logged indicating a failure, and the switch **102**, **104** is prevented from activation.

(48) The shift system **100** assists the vehicle **10** in executing gear shift requests by determining the relative safety based on the condition of the shifter driver interface **20**. While the shifter driver interface **20** is still operable with two active switches **102a**, **102b**, **104**, the shift system **100** provides a backup monitoring for potential failures by recording the diagnostic logs **214**, **216**. The incorporation of the interlock switch **104** minimizes the overall cost of the shift system **100** by utilizing a binary switch to identify the open state **110a** or closed state **110b** as it relates to a shift request. Further, as noted throughout, the binary switch minimizes the number of communication pathways through the use of a single resistor ladder circuit for the interlock switch **104**.

(49) A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. Accordingly, other implementations are within the scope of the following claims.

(50) The foregoing description has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular configuration are generally not limited to that particular configuration, but, where applicable, are interchangeable and can be used in a selected configuration, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

Claims

1. A vehicle shift system, comprising: a first gear switch operable between a plurality of gear states; a second gear switch operable between the plurality of gear states; an interlock switch operable

- between an open state and a closed state, the closed state corresponding to one of the plurality of gear states; and a controller communicatively coupled to each of the first gear switch, the second gear switch, and the interlock switch, the controller including a diagnostic counter and configured to register a diagnostic on the diagnostic counter when a signal from at least one of the first gear switch, the second gear switch, and the interlock switch is different from the other of the first gear switch, the second gear switch, and the interlock switch.
2. The vehicle shift system of claim 1, wherein the plurality of gear states includes at least one of a reverse state, a neutral state, and a drive state.
 3. The vehicle shift system of claim 2, wherein the closed state of the interlock switch corresponds to activation of at least one of the reverse state, the neutral state, and the drive state.
 4. The vehicle shift system of claim 1, wherein the controller includes a memory storing a predetermined diagnostic threshold, the controller configured to compare a signal from the interlock switch with a signal from each of the first gear switch and the second gear switch, respectively.
 5. The vehicle shift system of claim 4, wherein the controller is configured to detect one or more of the first gear switch, the second gear switch, and the interlock switch being activated via the respective signals.
 6. The vehicle shift system of claim 5, wherein the diagnostic counter of the controller is configured with a passing criteria and a non-passing criteria, the diagnostic counter configured to reduce a diagnostic log based on a predetermined number of pass accruals of the passing criteria.
 7. The vehicle shift system of claim 6, wherein the pass accruals include the controller receiving a signal from each of the first gear switch, the second gear switch, and the interlock switch.
 8. A shift system, comprising: a shifter including a first switch, a second switch, and an interlock switch and being operable between a plurality of gear states; and a controller communicatively coupled with the shifter to receive signals corresponding to each of the first switch, the second switch, and the interlock switch, the controller including a diagnostic counter that includes a passing criteria and a predetermined diagnostic threshold and being configured to adjust the diagnostic counter based on at least one of the signals received from the shifter being inconsistent with the others of the signals received.
 9. The shift system of claim 8, wherein the interlock switch is a resistor ladder circuit operable between an open state and a closed state.
 10. The shift system of claim 9, wherein the controller is configured to add a diagnostic to the diagnostic counter when a signal of the interlock switch is different than a signal of at least one of the first switch and the second switch.
 11. The shift system of claim 9, wherein the first switch and the second switch each includes the plurality of gear states, the plurality of gear states including at least one of a reverse state, a neutral state, and a drive state.
 12. The shift system of claim 11, wherein the closed state of the interlock switch includes at least one of the plurality of gear states.
 13. The shift system of claim 11, wherein the controller is configured to add a diagnostic to the diagnostic counter when the interlock switch is in the open state and the first switch and the second switch are in a same one of the plurality of gear states.
 14. The shift system of claim 8, wherein the diagnostic counter includes a standard diagnostic log and an interlock diagnostic log, the standard diagnostic log corresponding to signals received from the first switch and the second switch and the interlock diagnostic log corresponding to a signal received from the interlock switch.
 15. A shift system for a vehicle, comprising: a gear switch including a first switch and a second switch each operable between at least one of a reverse state, a neutral state, and a drive state; an interlock switch communicatively coupled to the gear switch and operable between an open state and a closed state, the closed state selectively corresponding to one of the reverse state, the neutral

state, and the drive state; and a controller communicatively coupled to the gear switch and the interlock switch and configured to receive signals from the interlock switch and the gear switch, the controller storing a diagnostic counter configured to track a difference between the signal received from the interlock switch and the signal received from the gear switch, the diagnostic counter including a predetermined diagnostic threshold for each of the gear switch and the interlock switch, respectively.

16. The shift system of claim 15, wherein the signal of the interlock switch includes one of the open state and the closed state of the interlock switch and the signals of the gear switch includes at least one of the reverse state, the neutral state, and the drive state of each of the first switch and the second switch.

17. The shift system of claim 16, wherein the controller is configured with a passing criteria and is configured to reduce a diagnostic count of the diagnostic counter based on the passing criteria.

18. The shift system of claim 17, wherein the passing criteria is defined by the interlock switch being in the closed state and both of the first switch and the second switch being in a same state of any one of the reverse state, the neutral state, and the drive state.

19. The shift system of claim 15, wherein the controller is configured to determine an activation time between the signals received from the gear switch and the interlock switch.

20. The shift system of claim 19, wherein the controller is configured to log a pass accrual based on the activation time.
